



**Bharati Vidyapeeth's
College of Engineering, New Delhi**

celebrating
Golden Jubilee
BHARATI VIDYAPEETH
Hon'ble Founder Dr. Patangrao Kadam

1964-2014



DISASTER MANAGEMENT

Department of Applied Sciences, BVCOE, New Delhi



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What is a Disaster?

A disaster is a sudden, calamitous event that seriously disrupts the functioning of a community or society and causes human, material, and economic or environmental losses that exceed the community's or society's ability to cope using its own resources.

$$(\underline{VULNERABILITY} + \underline{HAZARD}) / \underline{CAPACITY} = \underline{DISASTER}$$

Types of Disaster

- **Natural Disaster** : Natural disasters are extreme, sudden events caused by environmental factors that injure people and damage property. Earthquakes, windstorms, floods
- **Environmental emergency** : Disaster resulting from natural or human-induced factors, or a combination of these, that threatens to cause severe environmental damage.
- **Complex emergencies** : It combines internal conflict with large-scale displacements of people that may be due to food shortage, and failing economic, political, and social institutions.
 - **Pandemic emergencies** : Pandemic emergencies occurs when a highly infectious new disease emerges that humans have no immunity to. The difficulty responding to pandemics is the isolation they cause; normal emergency assistance is hampered by the need to limit exposure to infection.



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DISASTER MANAGEMENT: FLOODS, EARTHQUAKES, CYCLONES, LANDSLIDES

- The Indian subcontinent is very vulnerable to droughts, floods, cyclones, earthquakes, landslides, avalanches and forest fires. Among the 36 states and Union territories in the country, 22 are prone to disasters.
- Among all the disasters that occur in the country, floods are the most frequently occurring natural disasters, due to the irregularities of the Indian monsoon. About 75 percent of the annual rainfall in India is concentrated in three to four months of the monsoon season. As a result there is a very heavy discharge from the rivers during this period causing widespread floods. Approximately 40 million hectares of land in the country has been identified as being prone to floods. Major floods are mainly caused in the Ganga-Brahmaputra-Meghna basin which carries 60 percent of the total river flow of our country.



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- India has a long coastline of 5700 kms, which is exposed to tropical cyclones arising in the Bay of Bengal and the Arabian sea. The Indian Ocean is one of the six major cyclone prone regions of the world. In India, cyclones occur usually between April and May and also between October and December. The eastern coastline is more prone to cyclones as it is hit by about 80 percent of the total cyclones generated in the region.
- Droughts are a perennial feature in some states of India. Sixteen percent of the country's total area is drought prone. Drought is a significant environmental problem as it is caused by a lower than average rainfall over a long period of time. Most of the drought prone areas identified by the Government lie in the arid and semi-arid areas of the country.
- Earthquakes are considered to be one of the most destructive natural hazards. Most of the vulnerable areas are located in the Himalayan and sub-Himalayan regions.



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DISASTER MANAGEMENT

In simple terms, **Disaster Management** refers to how we can protect or preserve maximum number of lives and property during a natural disaster. But in a broader view, it can be defined as the organization and management of resources and responsibilities for dealing with all humanitarian aspects of emergencies, in particular preparedness, response and recovery in order to lessen the impact of disasters.



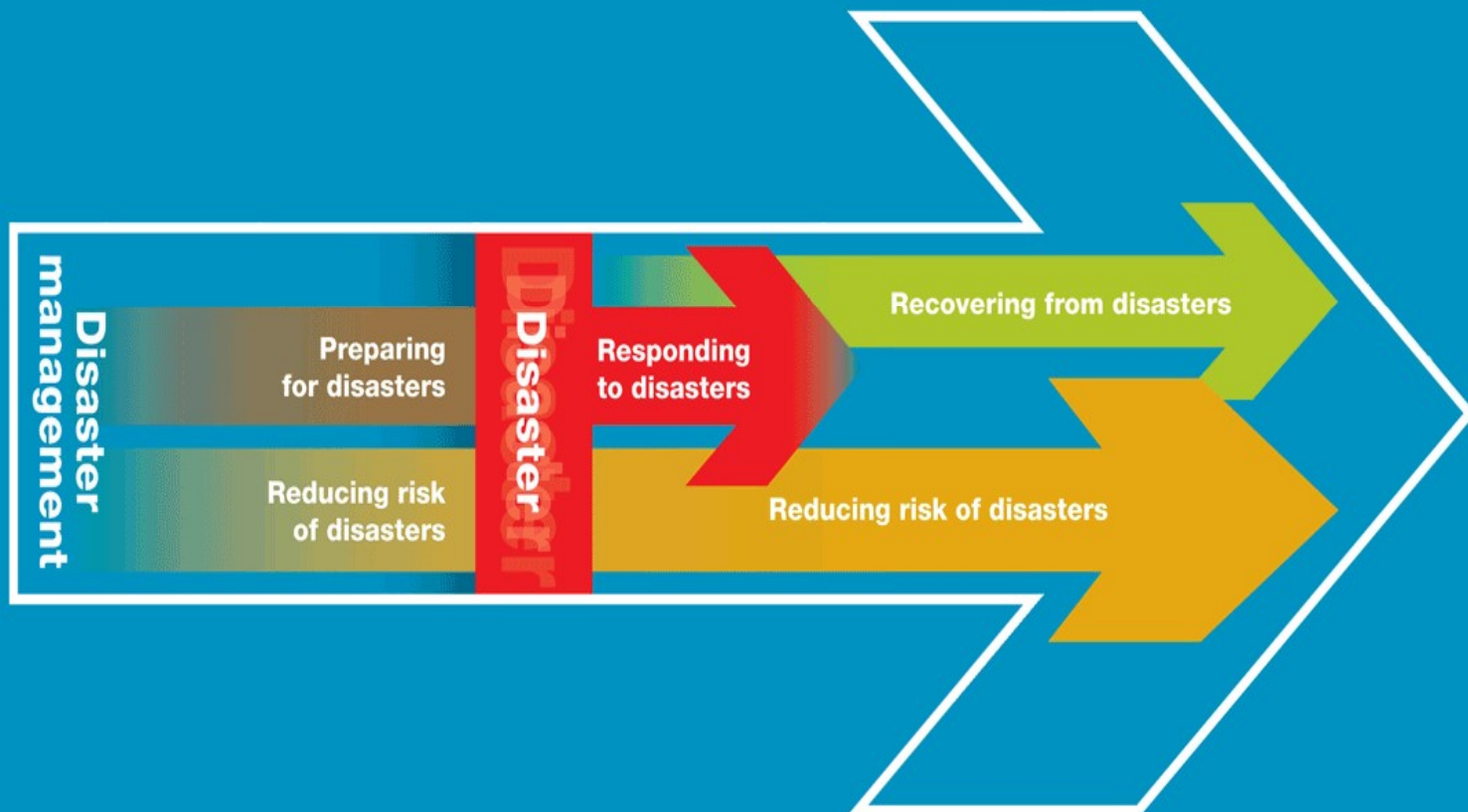
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FROM MANAGEMENT TO MITIGATION OF DISASTERS

Disaster management is a multidisciplinary area in which a wide range of issues that range from forecasting, warning, evacuation, search and rescue, relief, reconstruction and rehabilitation are included. It is also multi-sectoral as it involves administrators, scientists, planners, volunteers and communities. These roles and activities span the pre-disaster, during disaster and post disaster plans.



Prevention

Preparedness

**Disaster
Management
Cycle**

Recovery

Response



1.Response-Response measures are usually those which are taken immediately prior to and following disaster impact.

- Typical measures include :
- Implementation of plans
- Activation of the counter-disaster system
- Search and Rescue
- Provision of emergency food, shelter, medical assistance etc.
- Survey and assessment
- Evacuation measures



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2.Recovery-Three main categories of activity are normally regarded as coming within the recovery segment:

- Restoration
- Reconstruction
- Rehabilitation

3.Prevention & Mitigation-

Prevention : Action within this segment is designed to impede the occurrence of a disaster event and/or prevent such an occurrence having harmful effects on communities or key installations.

Mitigation : Action within this segment usually takes the form of specific programs intended to reduce the effects of disaster on a nation or community. For instance, some countries regard the development and application of building codes (which can reduce damage and loss in the event of earthquakes and cyclones) as being in the category of mitigation.



4.Preparedness-Examples of Preparedness measures are :

- The formulation & maintenance of valid, up-to-date counter-disaster plans
- Special provisions for emergency action
- The provisions of warning systems
- Emergency communications
- Public education and awareness
- Training programs, including exercises and tests.

FLOOD

➤ OCCURRENCE

As water falls onto the earth in the form of rain or snow, it seeps into the ground, but if the ground is frozen or the surface is impervious, that is, made of concrete or some other material which blocks water flow into the ground or the soil is already saturated then it cannot absorb water faster than it falls from the sky and this leads to the problem of FLOOD.

DIFFERENT TYPES OF FLOODS

1. **Fluvial or Riverine floods** – They occur when the water rises and overflows over the edges of a river or stream. This is the most common and can occur in any size channel, from small streams to huge rivers.



2. Flash floods : Flash flooding is characterized by an intense, high velocity torrent of water that occurs in an existing river channel with little or no notice . Flash floods are very dangerous and destructive not only because of the forces of the water, but also the hurtling debris that is often swept up in the flow.





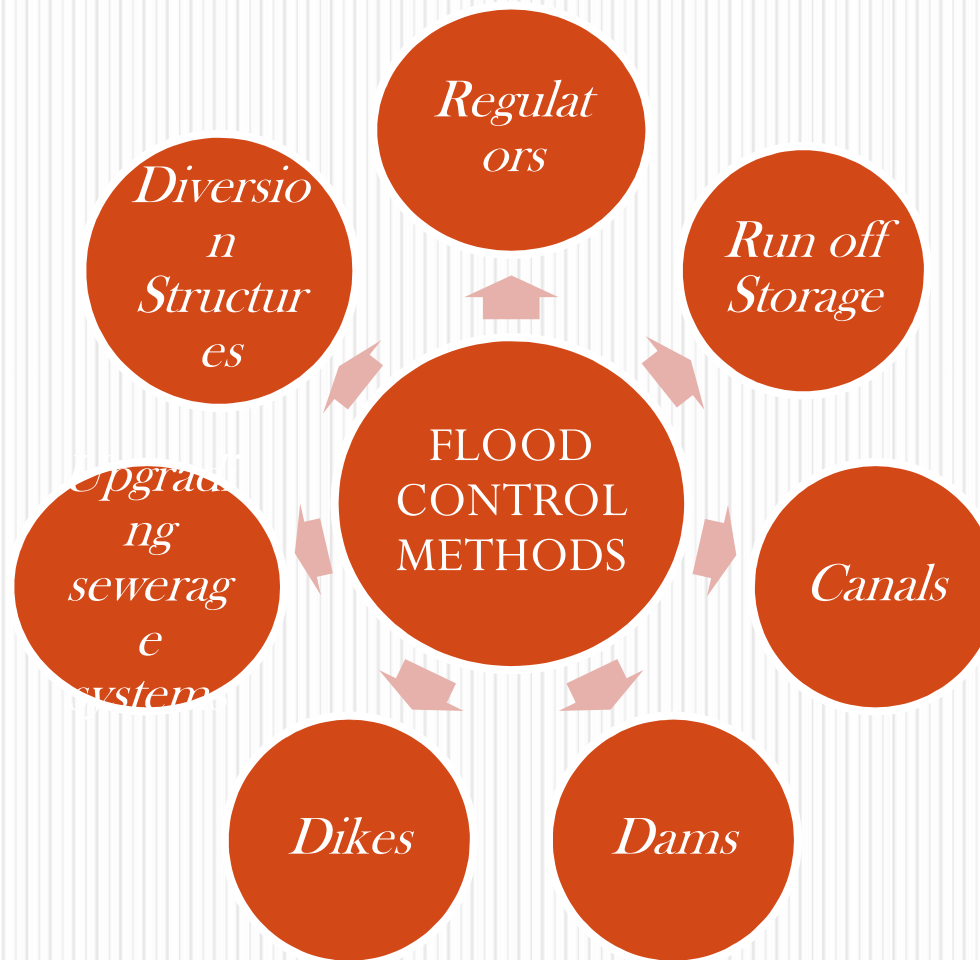
3. Coastal floods: Along the edges of oceans, and is driven predominantly by storm surges (waves significantly larger than normal) and wave damage. This kind of flooding is usually connected to hurricanes, tsunamis and tropical storms. The extent of coastal flooding is a function of the elevation inland flood waters penetrate which is controlled by the topography of the coastal land exposed to flooding.

4. Due to Engineering Issues: Flooding may also be caused by manmade issues as well as a weakly constructed dam could receive a more substantive battering than it was designed for, creating a flash flood in the regions downstream.

For example :- Kerala flood due to Idukki dam failure.



FLOOD CONTROL AND FLOOD RELIEF





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Preventions to be taken during a Flood

- Seek higher ground. Do not wait for instructions.
- Be aware of flash flood areas such as canals, streams, drainage channels.
- Be ready to evacuate.
- If instructed, turn off utilities at main switches and unplug appliances - do not touch electrical equipment if wet.
- If you must leave your home, do not walk through moving water. Six inches of moving water can knock you off your feet. Use a stick to test depth.
- Do not try to drive over a flooded road. If your car stalls, abandon it immediately and seek an alternate route.



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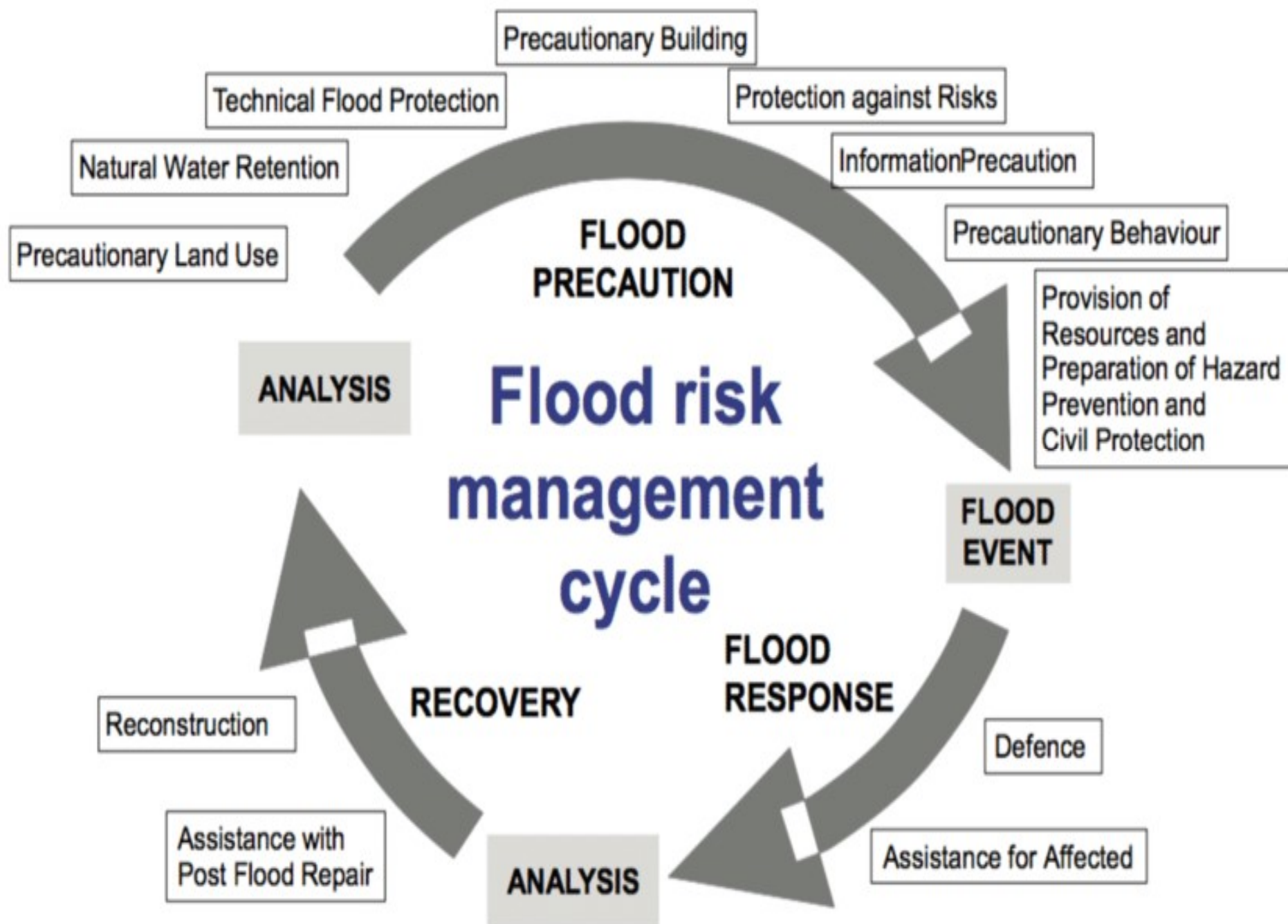
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Steps to be taken after a Flood

- Stay away from flood water - do not walk or drive through the area.
- Be aware of areas where water has receded. Roadways may have weakened and could collapse.
- Avoid downed power lines and muddy waters where power lines may have fallen.
- Do not drink tap water until advised by the Health Unit that the water is safe to drink.
- Once flood waters have receded you must not live in your home until the water supply has been declared safe for use, all flood-contaminated rooms have been thoroughly cleaned and disinfected, adequate toilet facilities are available, all electrical appliances and heating/cooling systems have been inspected, food, utensils and dishes have been examined, cleaned or disposed of, and floor drains and sumps have been cleaned and disinfected.



Flood Hazard Map

Flood Hazard Map (FHM) is defined as a map

- 1) with anticipated inundation area
- 2) with information for safe evacuation
- 3) made by municipality.

Flood hazard maps cover the geographical areas which could be flooded according to the following scenarios:

- a) floods with a low probability, or extreme event scenarios
- b) floods with a medium probability (likely return period ≥ 100 years)
- c) floods with a high probability, where appropriate.

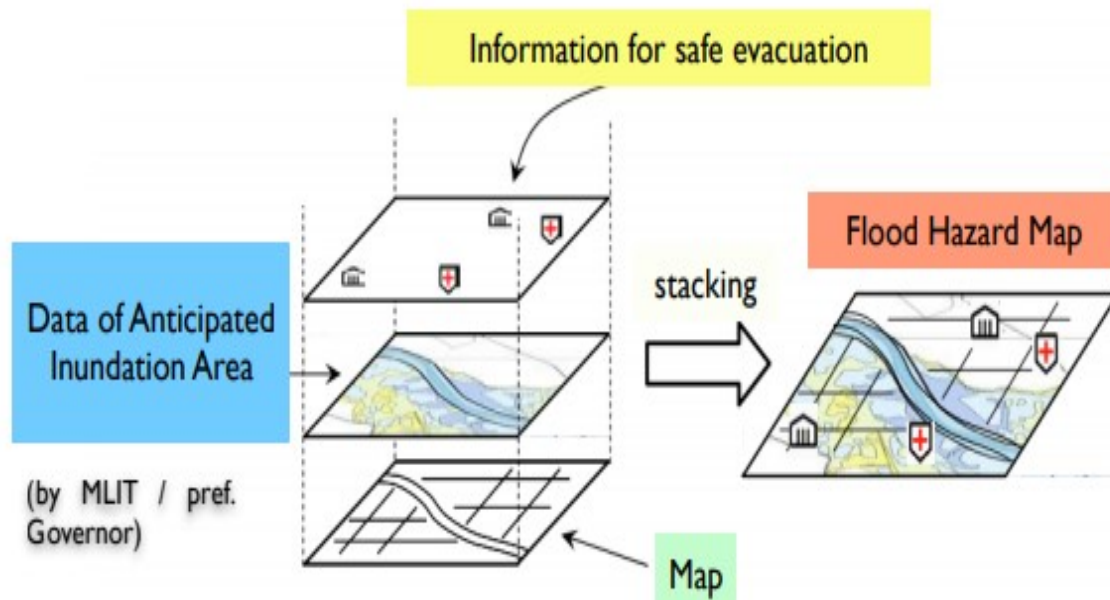
For each scenario the following elements shall be shown:

- a) the flood extent
- b) water depths or water level, as appropriate
- c) where appropriate, the flow velocity or the relevant water flow.

How to make FHM

The process of preparing FHM is as follows

- 1) The administrator of the river provides information on anticipated inundation area with depth.
- 2) Show the anticipated inundation area and depth on the topographical map and discuss necessary information for safe evacuation (by municipal office).
- 3) Stack above three kinds of information in a map (by municipal office).





*Case study: Kerala Floods
(2018)*



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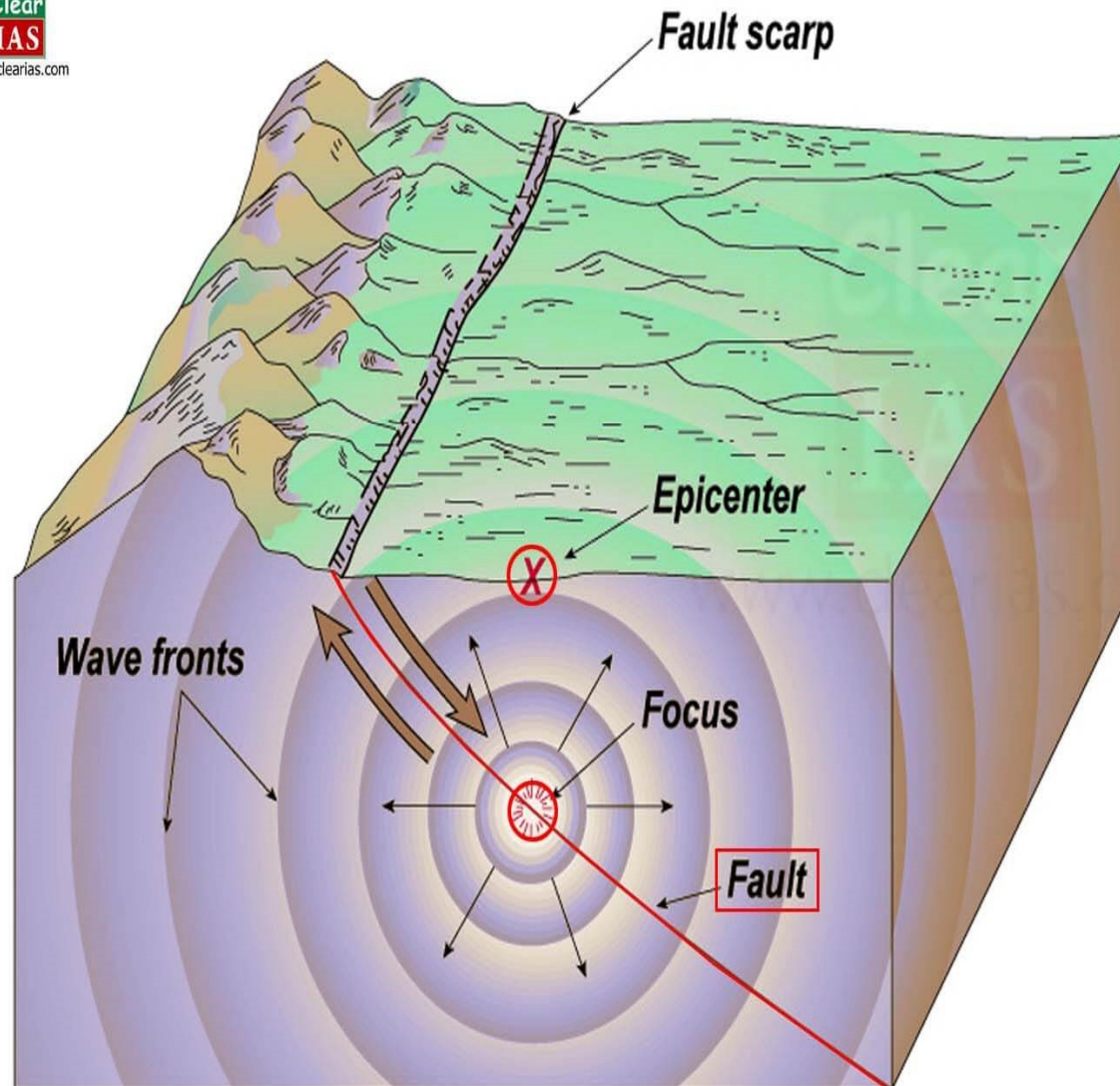
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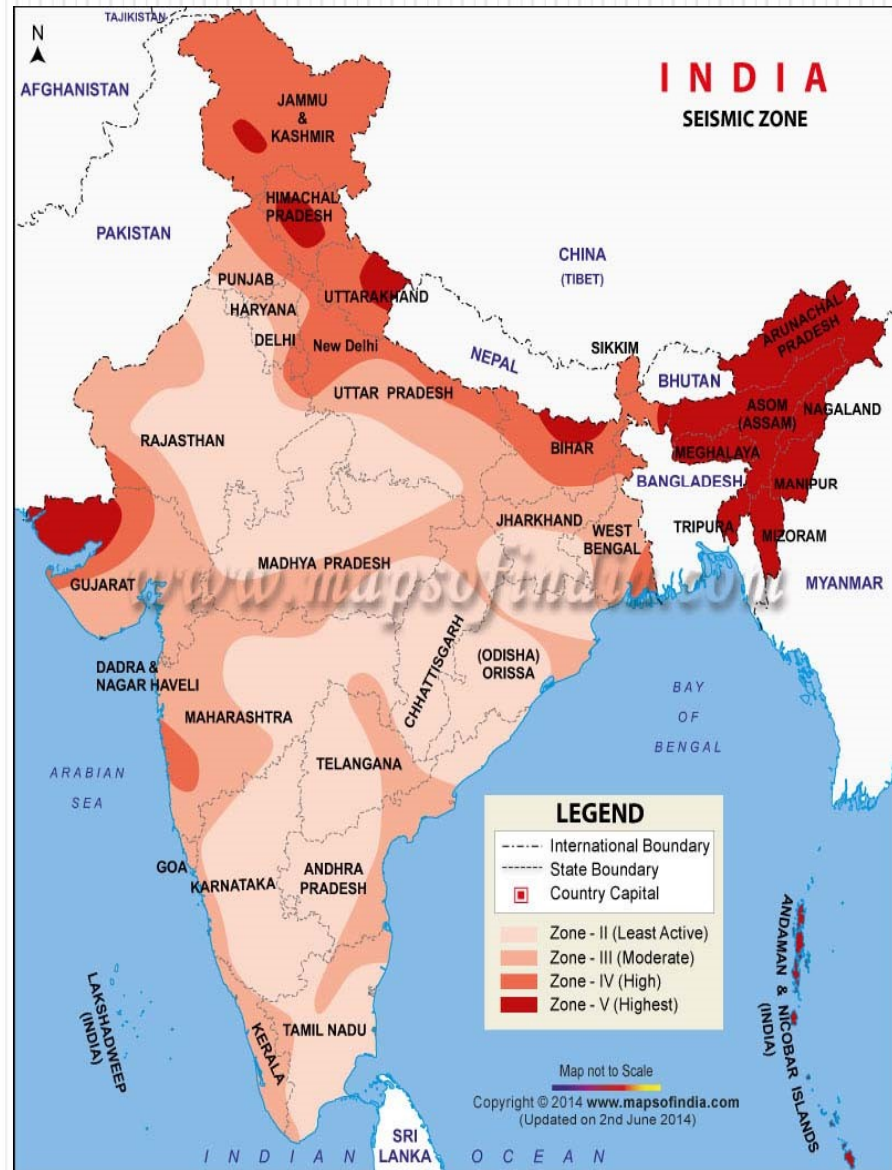
EARTHQUAKES

TERMS RELATED TO EARTHQUAKE

- **HYPOCENTRE/FOCUS** - The focus of an earthquake is where the earthquake originated from, usually underground on the fault zone.
- **EPICENTRE** - The epicentre of an earthquake is the point on the earth directly above the hypocentre.
- **MAGNITUDE** - The magnitude is a number that characterizes the relative size of an earthquake. Magnitude is based on measurement of the maximum motion recorded by a seismograph.
- **INTENSITY** - The intensity is a number describing the severity of an earthquake in terms of its effects on the earth's surface and on humans and their structures. Several scales exist, but the ones most commonly used are The Richter Scale and The Mercalli Scale.



SEISMIC ZONES IN INDIA



Why earthquakes occurs?

- Heat is being constantly produced in the interior of the earth . This heat develop convection currents in the molten rocks .
- As these currents rise, they cause fractures in the crust. The crust is divided into various segments (tectonic plates).
- These segments have been moving away from each other and also coming together, they collide with each other. The stress caused by collision leads to shaking of the earth's crust. We call this shaking and trembling of the earth's surface as an earthquake .
- Case study example: Gujarat earthquake 2001

EARTHQUAKE HAZARDS

Primary earthquake hazards are:

- Ground shaking
- Landslides
- Liquefaction – It is the mixing of sand or soil and **groundwater** during the shaking of a moderate or strong earthquake. When the water and soil are mixed, the ground becomes very soft and acts similar to quicksand.
- Surface rupture

Secondary earthquake hazards are those that are caused by the primary hazards, and may often be more catastrophic:

- Tsunami
- Seiche - Seismic seiches are standing waves set up on rivers, reservoirs, ponds, and lakes when seismic waves from an earthquake pass through the area.
- Flooding
- Fire

DISASTER MANAGEMENT STEPS FOR EARTHQUAKES

➤ **BEFORE** the earthquake

1. SECURE your space by identifying hazards and securing moveable items.
2. PLAN to be safe by creating a disaster plan and deciding how you will communicate in an emergency.
3. ORGANIZE disaster supplies in convenient locations.
4. MINIMIZE financial hardship by organizing important documents, strengthening your property, and considering insurance . **Survive and Recover**

➤ **DURING** the earthquake

1. DROP, COVER, and HOLD ON when the earth shakes.
2. IMPROVE safety after earthquakes by evacuating if necessary, helping the injured, and preventing further injuries or damage.

➤ **AFTER** the immediate threat of the earthquake has passed

RECONNECT and RESTORE daily life by reconnecting with others, repairing damage, and rebuilding community.



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CYCLONE

Cyclone refers to any inner spinning storm that rotates around a low-pressure centre. The low-pressure centre is also referred to as the 'eye' of the storm.

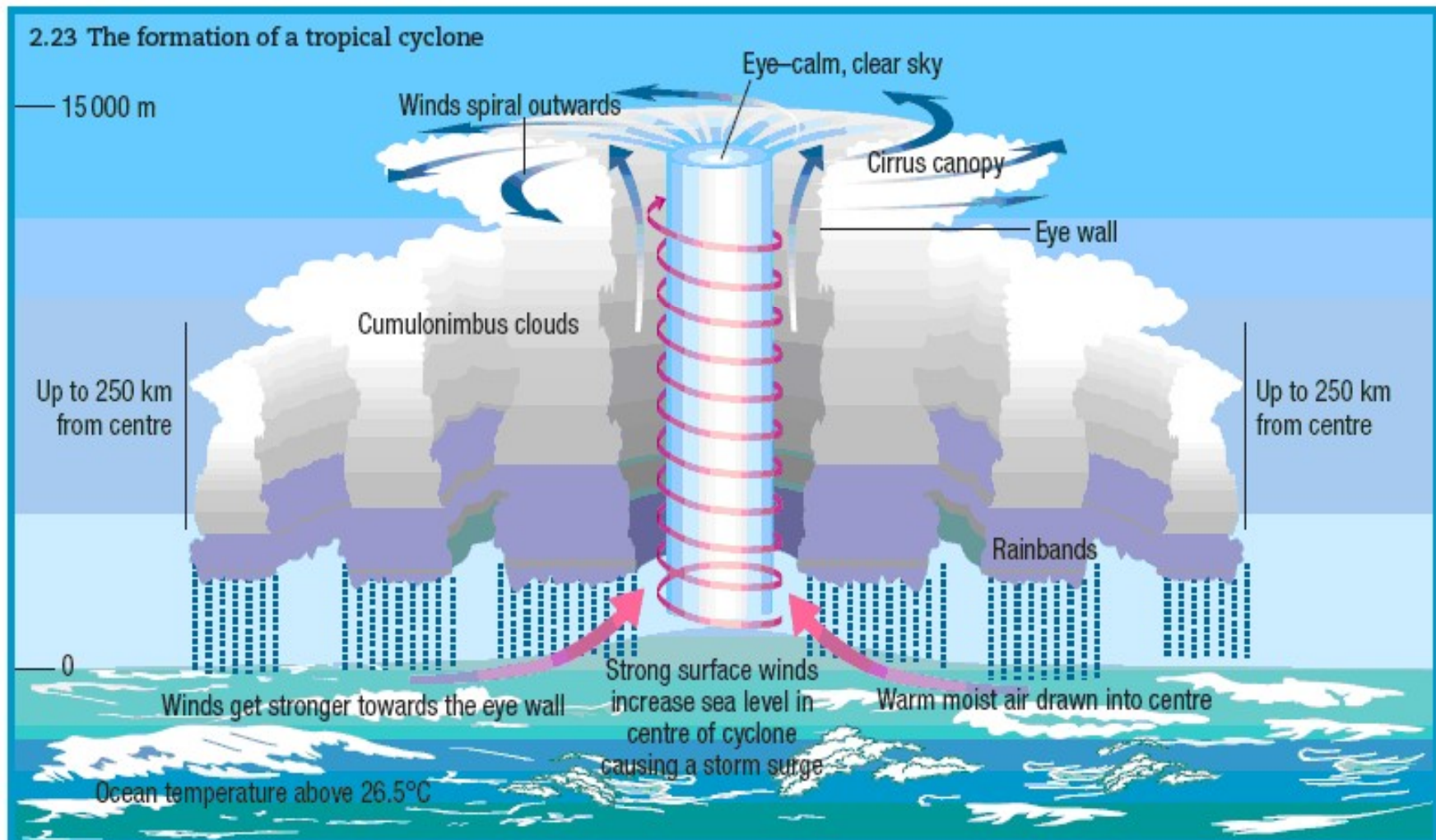
TYPES OF CYCLONES

- **Tropical Cyclone** – Cyclones that develop in the regions between the Tropic of Capricorn and Tropic of Cancer are called tropical cyclones.
- **Polar Cyclones** - Polar cyclones are cyclones that occur in polar regions like Greenland, Siberia and Antarctica.
- **Mesocyclone** - A mesocyclone is when part of a thunderstorm cloud starts to spin, which may eventually lead to a tornado.
- **Extratropical Cyclones** – The cyclones occur in temperature zones and high latitude regions.

How cyclones are formed

- ❑ Tropical cyclones form only over warm ocean waters near the equator.
- ❑ To form a cyclone, warm, moist air over the ocean rises upward from near the surface. As this air moves up and away from the ocean surface, it leaves less air near the surface. So basically as the warm air rises, it causes an area of lower air pressure below.
- ❑ Air from surrounding areas with higher air pressure pushes in to the low pressure area. Then this new “cool” air becomes warm and moist and rises, too. And the cycle continues...
- ❑ As the warmed, moist air rises and cools the water in the air forms clouds. The whole system of clouds and wind spins and grows, fed by the ocean's heat and water evaporating from the ocean surface.
- ❑ As the storm system rotates faster and faster, an eye forms in the centre. It is very calm and clear in the eye, with very low air pressure. Higher pressure air from above flows down into the eye

Structure of Cyclone



Destruction caused by cyclones

- Consequent strong winds. These, in turn generate storm surges.
- Abnormal rise of sea level near the coast caused by a severe tropical cyclone.
- Very strong winds may damage installations, dwellings, communication systems, trees ,etc.

Resulting in loss of life and property.

- Heavy rains due to cyclones may cause river floods.



CYCLONES – Do's & Don'ts

❑ Before the Cyclone season:

- Check the house, secure loose tiles, carry out repair works for doors and windows.
- Keep some dry non-perishable food always ready for emergency use.
- Keep some wooden boards ready so that glass windows can be boarded if needed.
- Remove dead woods or dying trees close to the house or any other things which can fly in strong winds.

❑ When the Cyclone starts:

- Listen to the radio about weather warnings.
- Keep monitoring the warnings. This will help you to prepare for a cyclone emergency.
- Pass on the information to others. Believe in the official information.
- Ignore rumours and do not spread them, this will help to avoid panic situations.



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Landslides



Landslides

- A landslide is the movement of rock and soil down a slope.
- Landslides range in size from a single boulder in a rock fall to a debris avalanche, which may have huge volumes of rock and soil capable of covering many kilometers.
- Landslides occur in a variety of environments, characterized by either steep or gentle slope gradients: from mountain ranges to coastal cliffs or even underwater, in which case they are called submarine landslides.

Causes for landslides

- Snow melting, or glaciers melting
- Rising of groundwater or increase of pore water pressure
- Volcanic eruptions
- Loss or absence of vertical vegetative structure, soil nutrients, and soil structure (e.g. after a wildfire – a fire in forests lasting for 3–4 days)
- Erosion of soil by rivers or ocean waves
- Deforestation, cultivation and construction
- Blasting and mining



Action Steps

- 1) Find out if there have been landslides in your area and where they might happen again.
- 2) Have an emergency evacuation plan and a kit of emergency supplies for your family
- 3) Check for signs that the ground might be moving:
 - a) New cracks or strange bulges in the ground, road, or footpaths
 - b) Leaning trees retaining walls or fences
 - c) Springs or waterlogged ground in areas that are not usually wet

What should you do if you think a landslide is about to happen ?





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What to do if you think a landslide is about to happen:

- 1) Evacuate and take your kit of emergency supplies and important documents.
- 2) Contact your local fire, police, or civil defence emergency management officer.
- 3) Warn neighbours who might be effected.
- 4) Do not return until the site has been inspected.

Emergency Management in case of any disaster



Emergency Management in case of any Disaster

You should be ready before a disaster happens. Start with these three steps:

1. Make up a kit of Survival Supplies

It must include:

- Food – Canned or dried food. Enough for three to four days
- Water – at least 9l per person in household
- Spare batteries, radio and torch
- First aid kit
- Essential medications
- Personal hygiene items – like toothbrush, soap, etc.





2. Make up an Emergency Plan for you and your Family

The emergency plan should answer the following questions:

- Where are your emergency supplies kept ?
- Where are the escape routes if a fire happens in the house ?
- Where are the safest places in your home to shelter in an earthquake ?
- Where is your nearest Civil Defence Center ?
- Where will you meet family members if an emergency happens during the day?
- Who has been approved to collect the children from school?



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NUCLEAR ACCIDENTS AND HOLOCAUST

A situation in which an uncontrollable chain reaction arises in a nuclear reactor.

Holocaust means destruction of life forms at mass level causing genocide, that is complete destruction of human civilization due to nuclear explosions.



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Causes of Nuclear Accidents

1. Natural disasters : due to tsunami and earthquake.
2. Improper following of procedures in a nuclear reaction
3. Faulty infrastructure of reactor due to which leakage of radioactive material
4. Malfunctioning of the system



Biological Effects

- Thyroid cancer: caused due to radioactive iodine
- Loss of white blood cells and platelets
- Leukemia: skin cancer caused due to exposure of gamma rays
- Bioaccumulation: of radioactive material in food chain.
- Genetic mutation

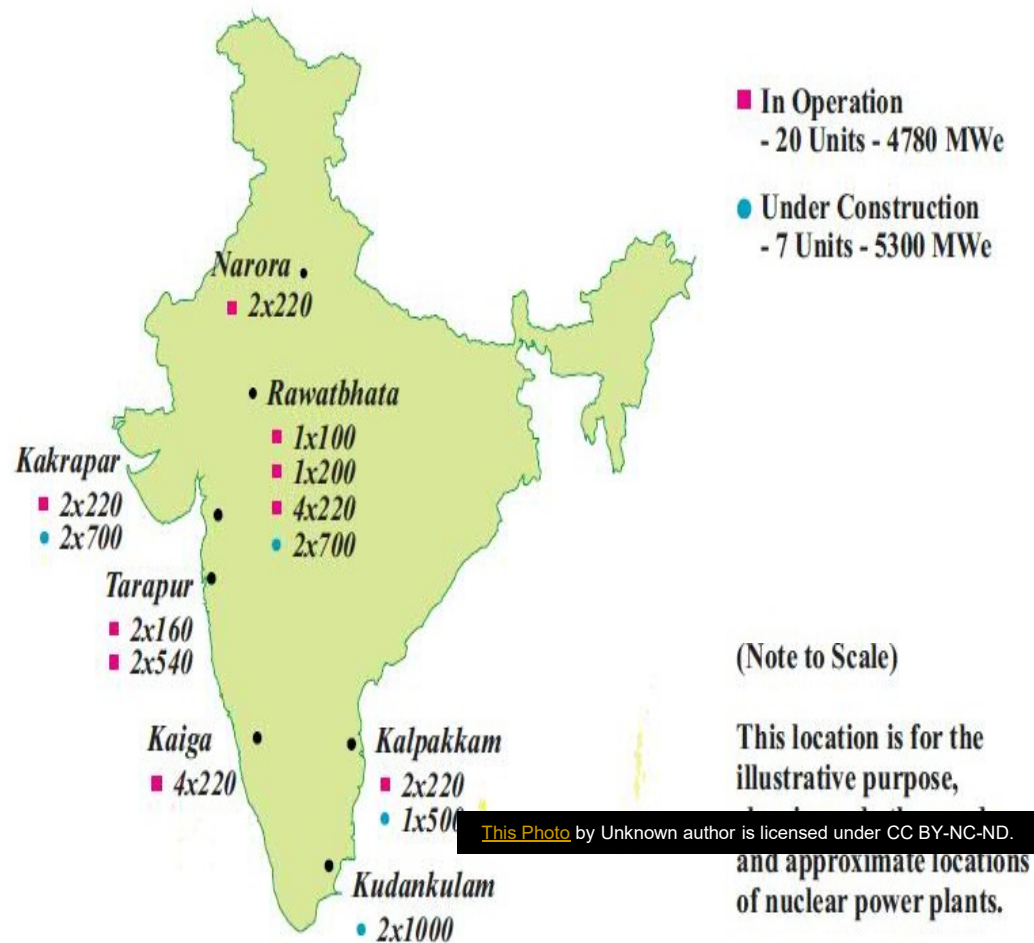
Non-Biological effects

- Loss of infrastructure
- Mass destruction of humans and other life forms.
- Immediate evacuation of people in affected areas.
- Cost of cleanup
- Harm to ozone layer
- Nuclear winters : The absorption of sunlight when large amounts of soot are injected into the atmosphere.



Is INDIA safe
from nuclear
accidents and
holocaust?

Let us see!



- India has 22 nuclear reactors in working which produces 30,292.91 GWh of electricity.
- India has approximately 110 nuclear weapons till today.
- World record production of fuel for indigenous nuclear reactors for 3 consecutive years 2014-15, 2015-16, and 2016-17.

INDIA as nuclear power



BUT!
Don't be so happy.

See Why?



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Nuclear incidents in India

Kalpakkam powerplant in
1987.

Kalpakkam powerplant in
2002.

Tarapur powerplant in 1989.

Tarapur powerplant
in 1992.

Main reasons behind these incidents



Negligence



Relying on older technology



Not following safety measures.



Employing unskilled professionals.



Untimely repairing of reactors.

A brief about Fukushima nuclear accident

- Fukushima Daichi nuclear accident was occurred in 2011 at Fukushima powerplant in northern Japan.
- The main reason behind this accident was tsunami occurred on on March 11, 2011 due to major earthquakes . Due to this the main generators of powerplant get damaged which cause breakage of cooling operation of nuclear reactors.
- As of year 2016, among those deaths, 1368 have been listed as "related to the nuclear power plant" according to media analysis.



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How susceptible is our India to this type of disaster?

- Most of our nuclear plants are in weak seismic zones but lie in coastal areas. Their structure is earthquake-resistant but they have not been tested against tsunami. The entire coastal region is believed to be vulnerable to tsunami.
- India is nuclear power state with a stockpile of fissile nuclear war heads and shares border with two other nuclear power states. Thus, the country is under a constant threat with regards to occurrence of both unintentional as well as intentional nuclear events.



- The risk from nuclear radiation accident in India is further increased by the region being endemic for iodine deficiency.
- India is also at a risk of nuclear theft such as terrorist can use this obtain nuclear bomb and harm us.
- India is ranked 19th in the nuclear theft according to report by NTI.



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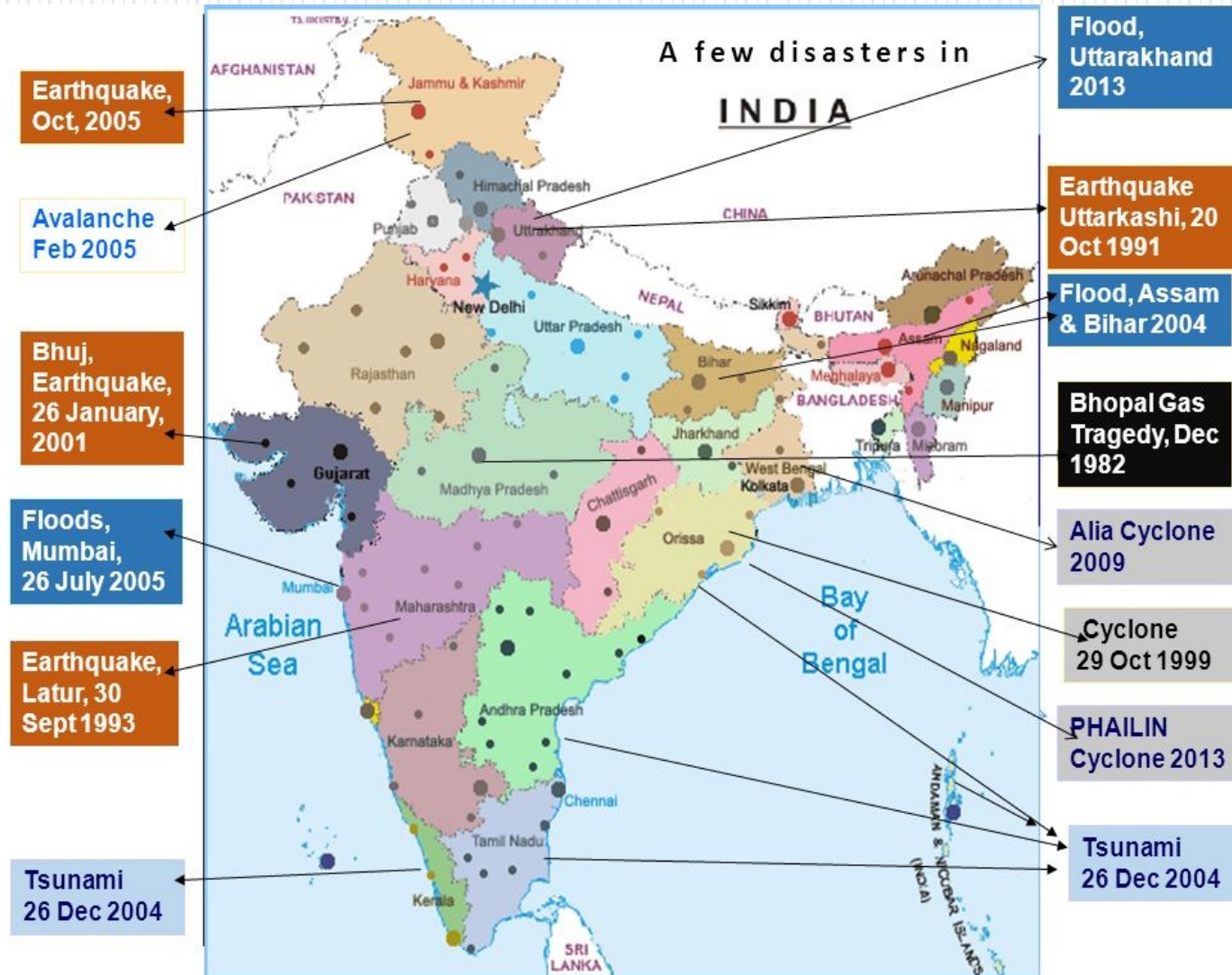
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Prevention of nuclear accidents

- Proper disposal of of radioactive wastes
- Awareness among public
- Proper transportation of radioactive materials
- Ban on nuclear weapons
- Making the violation of procedures a criminal offense

CASE STUDIES





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